

IMO 2026 — Problem 2

Solution by Claude Fable 5

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Source: `results/claude-fable-5/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

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Approaches tried

- **Power-of-a-point reformulation (successful).** Via the median-length formula, $OM = ON \iff \text{pow}(B, \omega) - \text{pow}(C, \omega) = \frac{1}{2}(AB^2 - AC^2)$, where ω is the circumcircle of AKL .
- **Auxiliary points X, Y (successful).** $Y := \text{ray } BK \cap AC$, $X := \text{ray } CL \cap AB$ (crossbar). The three angle hypotheses translate, via directed angles, into three concyclicities: B, X, Y, C on ω_1 ; C, M, K, X on ω_K ; B, N, L, Y on ω_L (with tangency of AB to ω_K at M in the degenerate case $X = M$, and of AC to ω_L at N when $Y = N$).
- **Affine power identity (the key step).** $F(Z) := \text{pow}(Z, \omega) - \text{pow}(Z, \omega_K) - \text{pow}(Z, \omega_L) + \text{pow}(Z, \omega_1)$ is an affine function of Z ; it vanishes at the three non-collinear points A, K, L , hence identically. Evaluating at B and C gives $\text{pow}(B, \omega) = \vec{BM} \cdot \vec{BX}$ and $\text{pow}(C, \omega) =$

$\vec{CN} \cdot \vec{CY}$; together with $AB \cdot AX = AC \cdot AY$ (power of A w.r.t. ω_1) and the median formula this yields $OM = ON$.

- **Dead end avoided:** first tried radical-axis arguments with the second intersection points K^*, L^* of line BY with ω_K and of CX with ω_L ; it works but requires awkward tangency case analysis. The affine function F replaces all of it.
- Numerical verification: `code/verify.py`, `code/check_signs.py`, `code/verify2.py` confirm every claim (several triangles; generic case; both tangency cases $X = M$ and $Y = N$; all signed-angle sign claims; $F \equiv 0$; $OM = ON$) to machine precision.

Current best

Complete rigorous proof below. Core of the argument: with $X = \text{ray } CL \cap AB$, $Y = \text{ray } BK \cap AC$, the hypotheses force B, X, Y, C, C, M, K, X and B, N, L, Y to be concyclic; then the affine function $\text{pow}(\cdot, \omega) - \text{pow}(\cdot, \omega_K) - \text{pow}(\cdot, \omega_L) + \text{pow}(\cdot, \omega_1)$ vanishes at A, K, L , hence everywhere; evaluating at B, C and using $AB \cdot AX = AC \cdot AY$ and the median-length formula gives $OM = ON$.

Full proof

0. Conventions and notation

We identify the plane with $\mathbb{C}$. For $u, v \in \mathbb{C}$ write

$$\langle u, v \rangle = \text{Re}(\bar{u}v), \quad u \times v = \text{Im}(\bar{u}v),$$

the usual dot and cross products of plane vectors.

Unsigned angles. For points $P, Q \neq V$, $\angle PVQ \in [0^\circ, 180^\circ]$ is the ordinary (unsigned) angle, i.e. $\angle PVQ = |\arg \frac{Q-V}{P-V}|$ with $\arg \in (-180^\circ, 180^\circ]$.

Signed ray angles. For nonzero u, v let $\angle(u \rightarrow v) := \arg \frac{v}{u} \in (-180^\circ, 180^\circ]$. For rays from V through P and through Q put $\angle(VP \rightarrow VQ) := \angle((P-V) \rightarrow (Q-V))$. Since $\frac{v}{u} = \frac{\bar{u}v}{|u|^2}$ we have:

(0.1) $|\angle(u \rightarrow v)| = \angle(u, v)$ (the unsigned angle), and if $u \times v \neq 0$ then the sign of $\angle(u \rightarrow v)$ equals the sign of $u \times v$. Moreover $\angle(u \rightarrow w) \equiv \angle(u \rightarrow v) + \angle(v \rightarrow w) \pmod{360^\circ}$ and $\angle(v \rightarrow u) \equiv -\angle(u \rightarrow v) \pmod{360^\circ}$ (because $\arg$ is additive mod 360° under multiplication and $\arg(u/v) = -\arg(v/u) \pmod{360^\circ}$).

Directed line angles. For lines ℓ_1, ℓ_2 with direction vectors $d_1, d_2 \neq 0$ set

$$\angle(\ell_1, \ell_2) := \arg \frac{d_2}{d_1} \pmod{180^\circ}.$$

This is well defined: replacing d_i by a nonzero real multiple changes $\arg \frac{d_2}{d_1}$ by 0° or 180° . It is additive mod 180° , and $\angle(\ell_1, \ell_2) \equiv 0$ iff $\ell_1 \parallel \ell_2$ (including $\ell_1 = \ell_2$). If rays r_1, r_2 span the lines ℓ_1, ℓ_2 , then $\angle(\ell_1, \ell_2) \equiv \angle(r_1 \rightarrow r_2) \pmod{180^\circ}$. For points we write $\angle(PQ, RS)$ for the directed angle from line PQ to line RS .

Power of a point. For a circle Γ with centre Z_0 and radius $\rho > 0$ and a point Z , $\text{pow}(Z, \Gamma) := |Z - Z_0|^2 - \rho^2$.

Sides of a line. For a line through P with direction d , points Z_1, Z_2 are *strictly on the same side* iff $d \times (Z_1 - P)$ and $d \times (Z_2 - P)$ are nonzero of equal sign; *strictly opposite sides* iff nonzero of opposite signs.

Interior of a triangle / of an angle. For an affinely independent triple P, Q, R , the (open) interior of triangle PQR is $\{\alpha P + \beta Q + \gamma R : \alpha, \beta, \gamma > 0, \alpha + \beta + \gamma = 1\}$; “inside triangle PQR ” means lying in this set. For $0^\circ < \angle QVR < 180^\circ$, “ P lies inside angle QVR ” means: P is strictly on the same side of line VQ as R and strictly on the same side of line VR as Q .

Throughout, $b := AC = |C - A|$, $c := AB = |B - A|$, and

$$\theta := \angle KBA = \angle ACL, \quad \varphi := \angle LBK = \angle LNC, \quad \psi := \angle LCK = \angle BMK,$$

ω denotes the circumcircle of triangle AKL and O its centre, $R_\omega = |OA|$ its radius. Since the problem speaks of *triangle* AKL and its circumcentre, the points A, K, L are pairwise distinct and not collinear; this is part of the hypothesis.

Reflections, rotations and translations of the plane preserve all hypotheses and the conclusion, so **we may and do assume that ABC is positively oriented**:

$$\delta := (B - A) \times (C - A) > 0. \quad (0.2)$$

1. Preliminary lemmas

Lemma 1 (line–circle intersections and power). Let Γ be a circle with centre Z_0 , radius $\rho > 0$, and let ℓ be a line. (i) $\ell \cap \Gamma$ has at most two points. (ii) If $M \in \Gamma$, there is exactly one line through M meeting Γ only at M (the **tangent** at M); its direction is $i(M - Z_0)$, i.e. it is the line through M perpendicular to MZ_0 . (iii) If $Z \in \ell$ and ℓ meets Γ at U and V — where in the tangent case we set $U = V$ — then

$$\text{pow}(Z, \Gamma) = \langle U - Z, V - Z \rangle,$$

the product of signed lengths $ZU \cdot ZV$ along ℓ .

Proof. Parametrize $\ell = \{Z + td : t \in \mathbb{R}\}$ with $|d| = 1$, $Z \in \ell$ arbitrary. Then

$$|Z + td - Z_0|^2 - \rho^2 = t^2 + 2t \langle d, Z - Z_0 \rangle + \text{pow}(Z, \Gamma), \quad (1.1)$$

a monic real quadratic in t ; it has at most two roots, proving (i). For (ii): take $Z = M \in \Gamma$; then $\text{pow}(M, \Gamma) = 0$ and the roots are $t = 0$ and $t = -2\langle d, M - Z_0 \rangle$; they coincide iff $\langle d, M - Z_0 \rangle = 0$, i.e. iff $d \perp (M - Z_0)$, i.e. iff d is a real multiple of $i(M - Z_0)$; that determines a unique line through M . For (iii): if $U = Z + t_1 d$, $V = Z + t_2 d$ (with $t_1 = t_2$ in the tangent case, when (1.1) has a double root), then by Vieta $t_1 t_2 = \text{pow}(Z, \Gamma)$, and $\langle U - Z, V - Z \rangle = t_1 t_2 \langle d, d \rangle = t_1 t_2$. $\square$

Lemma 2 (directed inscribed-angle criterion). Let P, Q, R be pairwise distinct, non-collinear points and let Γ be the (unique) circle through them. Then for every point $S \notin \{P, Q\}$:

$$S \in \Gamma \iff \angle(SP, SQ) \equiv \angle(RP, RQ) \pmod{180^\circ}.$$

Proof. Uniqueness of Γ is classical (its centre is the common point of the perpendicular bisectors of PQ and PR , which are not parallel).

Put $w := (Q - R)\overline{(P - R)} \neq 0$. Since $\frac{Q-R}{P-R} = \frac{w}{|P-R|^2}$ and P, Q, R are not collinear, $\arg \frac{Q-R}{P-R} \not\equiv 0 \pmod{180^\circ}$, i.e. $w \notin \mathbb{R}$, so $\text{Im } w \neq 0$.

Define $g : \mathbb{C} \rightarrow \mathbb{R}$ by $g(Z) := \operatorname{Im} [w(P - Z)\overline{(Q - Z)}]$. Expanding $(P - Z)\overline{(Q - Z)} = P\bar{Q} - P\bar{Z} - \bar{Q}Z + |Z|^2$ shows

$$g(Z) = (\operatorname{Im} w)|Z|^2 + \lambda(Z)$$

with λ a real affine function of $Z \in \mathbb{R}^2$. Hence $\frac{1}{\operatorname{Im} w} g(Z) = |Z - Z_1|^2 - \rho_1$ for some $Z_1 \in \mathbb{C}, \rho_1 \in \mathbb{R}$, so the zero set $\{g = 0\}$ is empty, a single point, or a circle. Now

$$g(P) = 0, \quad g(Q) = 0, \quad g(R) = \operatorname{Im} [(Q - R)\overline{(P - R)}(P - R)\overline{(Q - R)}] = \operatorname{Im} [|P - R|^2 |Q - R|^2] = 0.$$

So $\{g = 0\}$ contains three distinct non-collinear points, hence it is a circle through P, Q, R ; by uniqueness $\{g = 0\} = \Gamma$.

Finally, fix $S \notin \{P, Q\}$. All four numbers $Q - S, P - S, Q - R, P - R$ are nonzero, and

$$\angle(SP, SQ) \equiv \angle(RP, RQ) \pmod{180^\circ} \iff \arg \left[\frac{Q-S}{P-S} \right] \equiv \arg \left[\frac{Q-R}{P-R} \right] \pmod{180^\circ} \iff \frac{Q-S}{P-S} \cdot \overline{\left(\frac{Q-R}{P-R} \right)} \in \mathbb{R} \setminus \{0\}.$$

Multiplying by the positive real $|P - S|^2 |P - R|^2$, the last condition is equivalent to $(Q - S)\overline{(P - S)} \cdot \overline{(Q - R)}(P - R) \in \mathbb{R}$, i.e. (taking conjugates) to $w(P - S)\overline{(Q - S)} \in \mathbb{R}$, i.e. to $g(S) = 0$, i.e. to $S \in \Gamma$. $\square$

Lemma 3 (directed tangent–chord). Let Γ be a circle through the pairwise distinct points M, K, C , and let t be the tangent to Γ at M . Then (i) $\angle(t, MK) \equiv \angle(CM, CK) \pmod{180^\circ}$; (ii) if a line ℓ through M satisfies $\angle(\ell, MK) \equiv \angle(CM, CK) \pmod{180^\circ}$, then $\ell = t$; in particular ℓ meets Γ only at M .

Proof. (i) Maps $Z \mapsto aZ + \beta$ ($a \neq 0$) preserve directed line angles (all direction vectors get multiplied by a , so ratios of directions are unchanged), map circles to circles and tangent lines to tangent lines (they are bijections preserving the intersection pattern). Hence we may assume Γ is the unit circle centred at 0. Write $m, k, c \in \Gamma$, so $|m| = |k| = |c| = 1, \bar{m} = 1/m$ etc. By Lemma 1(ii), t has direction im . Consider

$$z := \frac{(k - m)(m - c)}{m(k - c)}.$$

Then

$$\bar{z} = \frac{(\bar{k} - \bar{m})(\bar{m} - \bar{c})}{\bar{m}(\bar{k} - \bar{c})} = \frac{\frac{m-k}{km} \cdot \frac{c-m}{mc}}{\frac{1}{m} \cdot \frac{c-k}{kc}} = \frac{(m-k)(c-m)}{m(c-k)} = -z,$$

so z is purely imaginary; also $z \neq 0$. Hence $z/i \in \mathbb{R} \setminus \{0\}$. Now

$$\angle(t, MK) - \angle(CM, CK) \equiv \arg \frac{k - m}{im} - \arg \frac{k - c}{m - c} = \arg \left[\frac{(k - m)(m - c)}{im(k - c)} \right] = \arg \frac{z}{i} \equiv 0 \pmod{180^\circ}.$$

(ii) From (i), $\angle(\ell, MK) \equiv \angle(t, MK)$, hence $\angle(\ell, t) \equiv 0$, so $\ell \parallel t$; both pass through M , so $\ell = t$. $\square$

Lemma 4 (median-length formula). For any points O, P, Q , if $W = \frac{1}{2}(P + Q)$ then

$$4|OW|^2 = 2|OP|^2 + 2|OQ|^2 - |PQ|^2.$$

Proof. With vectors from O : $4|W - O|^2 = |(P - O) + (Q - O)|^2 = |P - O|^2 + 2\langle P - O, Q - O \rangle + |Q - O|^2$ and $|PQ|^2 = |P - O|^2 - 2\langle P - O, Q - O \rangle + |Q - O|^2$. Add. $\square$

Lemma 5 (angle addition inside an angle). Let V, Q, R be points with $0^\circ < \angle QVR < 180^\circ$ and let P lie inside angle QVR . Then $\angle QVP + \angle PVR = \angle QVR$, and $0^\circ < \angle QVP < \angle QVR$.

Proof. Unsigned angles and sides of lines are invariant under translations, rotations and reflections, so we may place $V = 0$ and assume the unit vector along VQ is $q = 1$ and $\gamma := \angle QVR \in (0^\circ, 180^\circ)$ with unit vector $r = (\cos \gamma, \sin \gamma)$ along VR (reflect in the x -axis if needed). Let $p = (\cos \sigma, \sin \sigma)$, $\sigma \in (-180^\circ, 180^\circ]$, be the unit vector along VP . “ P on the same side of line VQ (the x -axis) as R ” means $\sin \sigma > 0$, so $\sigma \in (0^\circ, 180^\circ)$. “ P on the same side of line VR as Q ” means $r \times p$ and $r \times q$ have the same sign; $r \times q = -\sin \gamma < 0$ and $r \times p = \sin(\sigma - \gamma)$; since $\sigma - \gamma \in (-180^\circ, 180^\circ)$, the condition $\sin(\sigma - \gamma) < 0$ means $\sigma < \gamma$. So $0 < \sigma < \gamma$, and then $\angle QVP = \sigma$, $\angle PVR = \gamma - \sigma$, $\angle QVR = \gamma$. $\square$

2. Position facts

Write points as affine combinations $Z = \alpha_Z A + \beta_Z B + \gamma_Z C$ with $\alpha_Z + \beta_Z + \gamma_Z = 1$ (barycentric coordinates w.r.t. ABC ; not to be confused with the side lengths b, c). From $Z - A = \beta_Z(B - A) + \gamma_Z(C - A)$, $Z - B = \alpha_Z(A - B) + \gamma_Z(C - B)$, $Z - C = \alpha_Z(A - C) + \beta_Z(B - C)$ and the cyclic identity

$$(B - A) \times (C - A) = (C - B) \times (A - B) = (A - C) \times (B - C) = \delta \quad (2.1)$$

(direct expansion of all three gives $A \times B + B \times C + C \times A$), we get

$$(B - A) \times (Z - A) = \gamma_Z \delta, \quad (C - B) \times (Z - B) = \alpha_Z \delta, \quad (A - C) \times (Z - C) = \beta_Z \delta. \quad (2.2)$$

In particular, by (0.2): Z is strictly on the same side of line AB as C iff $\gamma_Z > 0$; strictly on the same side of BC as A iff $\alpha_Z > 0$; strictly on the same side of CA as B iff $\beta_Z > 0$. The interior of triangle ABC is exactly $\{\alpha_Z, \beta_Z, \gamma_Z > 0\}$.

(P1) K lies strictly inside triangle ABC and $K \notin$ lines AB, BC, CM . Indeed $K = \alpha B + \mu M + \gamma C$ with $\alpha, \mu, \gamma > 0$, $\alpha + \mu + \gamma = 1$ (B, M, C are affinely independent since M is an interior point of AB and $C \notin AB$). As $M = \frac{1}{2}(A + B)$,

$$K = \frac{\mu}{2}A + (\alpha + \frac{\mu}{2})B + \gamma C,$$

all coefficients positive, so K is interior to ABC . In coordinates w.r.t. B, M, C : line $BM = AB$ is $\{\gamma = 0\}$, line BC is $\{\mu = 0\}$, line CM is $\{\alpha = 0\}$; the interior has $\alpha, \mu, \gamma > 0$.

(P2) L lies strictly inside triangle ABC and $L \notin$ lines AC, BC, BN . Same argument with $L = \alpha' B + \nu N + \gamma' C$, $N = \frac{1}{2}(A + C)$: $L = \frac{\nu}{2}A + \alpha' B + (\gamma' + \frac{\nu}{2})C$; line $NC = AC$ is $\{\alpha' = 0\}$, line BC is $\{\nu = 0\}$, line BN is $\{\gamma' = 0\}$.

In particular $A \neq K$, $A \neq L$ (A is not interior), $K \notin AB$, $L \notin AC$, so

$$0^\circ < \theta = \angle KBA < 180^\circ, \quad 0^\circ < \varphi = \angle LNC < 180^\circ, \quad 0^\circ < \psi = \angle BMK < 180^\circ. \quad (2.3)$$

(P3) Definition and position of Y and X . The ray from B through K meets segment AC at a point Y with $Y \neq A, C$; the ray from C through L meets segment AB at a point X with $X \neq A, B$.

Proof for Y. Let K have barycentric coordinates $(\alpha_K, \beta_K, \gamma_K)$, all positive, by (P1). First, A and C are strictly on opposite sides of line BK (direction $K - B$):

$$(K-B) \times (A-B) = -(A-B) \times (K-B), \quad (A-B) \times (K-B) = (A-B) \times (K-A) = -(B-A) \times (K-A) = -\gamma_K \delta < 0$$

so $(K-B) \times (A-B) = \gamma_K \delta > 0$; and by (2.2), $(K-B) \times (C-B) = -(C-B) \times (K-B) = -\alpha_K \delta < 0$. (Here $(A-B) \times (K-B) = (A-B) \times (K-A)$ because $(A-B) \times (A-B) = 0$.) Hence segment AC meets line BK in exactly one point Y , and $Y \neq A, C$ (strict signs), $Y \neq B$ (as $B \notin AC$, else $\delta = 0$).

Next, Y lies on the ray from B through K . Write $Y = B + t(K - B)$, $t \neq 0$. On one hand, Y has barycentric coordinates $(1-s, 0, s)$ with $s \in (0, 1)$, so by (2.2) $(B-A) \times (Y-A) = s\delta > 0$. On the other hand $Y-A = (B-A) + t(K-B)$ and $(B-A) \times (K-B) = (B-A) \times (K-A) = \gamma_K \delta$, so $(B-A) \times (Y-A) = t\gamma_K \delta$. Comparing, $t > 0$.

Proof for X. Symmetric. Let L have barycentric coordinates $(\alpha_L, \beta_L, \gamma_L)$, all positive. Then $(L-C) \times (A-C) = -(A-C) \times (L-C) = -\beta_L \delta < 0$ by (2.2), while from $L-C = \alpha_L(A-C) + \beta_L(B-C)$ and (2.1), $(L-C) \times (B-C) = \alpha_L(A-C) \times (B-C) = \alpha_L \delta > 0$. So A, B are strictly on opposite sides of line CL , and segment AB meets line CL in a single point $X \neq A, B$. Writing $X = C + t(L-C)$, and noting that X has barycentric coordinates $(1-s, s, 0)$ with $s \in (0, 1)$, (2.2) gives $(A-C) \times (X-C) = \beta_X \delta = s\delta > 0$, while $(A-C) \times (X-C) = t(A-C) \times (L-C) = t\beta_L \delta$; hence $t > 0$. $\square$

Introduce the signed coordinates along the two sides:

$$x := AX \in (0, c), \quad y := AY \in (0, b),$$

so that on the line AB (coordinatized from A towards B) the points A, X, M, B sit at $0, x, \frac{c}{2}, c$, and on line AC the points A, Y, N, C sit at $0, y, \frac{b}{2}, b$.

Note for later use: since Y lies on ray BK and $Y \neq B$, **line** $BY = \mathbf{line} BK$; since X lies on ray CL and $X \neq C$, **line** $CX = \mathbf{line} CL$. Also $X \neq Y$ (lines AB and AC meet only at A , and $X \neq A$).

3. Translating the hypotheses into directed angles

By (P1)–(P2), K and L are interior to ABC ; let their barycentric coordinates be positive. We use (2.2) and (0.1) repeatedly. Below, “ Z is on the C -side of AB ” abbreviates “strictly on the same side of line AB as C ”, etc.

(S1) If Z is on the C -side of AB (i.e. $\gamma_Z > 0$), then $\sphericalangle(BA \rightarrow BZ) = -\angle ABZ$ and $\sphericalangle(MB \rightarrow MZ) = +\angle BMZ$. *Proof.* $(A-B) \times (Z-B) = -(B-A) \times (Z-B) = -(B-A) \times (Z-A) = -\gamma_Z \delta < 0$, so by (0.1) the signed angle $\sphericalangle(BA \rightarrow BZ)$ is negative, of absolute value $\angle ABZ$. Also $B-M = \frac{1}{2}(B-A)$ and $(B-A) \times (Z-M) = (B-A) \times (Z-A) - (B-A) \times (M-A) = \gamma_Z \delta - 0 > 0$, so $\sphericalangle(MB \rightarrow MZ) = +\angle BMZ$. $\square$

(S2) If Z is on the B -side of AC (i.e. $\beta_Z > 0$), then $\sphericalangle(CA \rightarrow CZ) = +\angle ACZ$ and $\sphericalangle(NC \rightarrow NZ) = -\angle CNZ$. *Proof.* By (2.2), $(A-C) \times (Z-C) = \beta_Z \delta > 0$, so by (0.1) $\sphericalangle(CA \rightarrow CZ) = +\angle ACZ$. For the second claim: $C-N = \frac{1}{2}(C-A)$, and since $N-A$ is parallel to $C-A$,

$$(C-N) \times (Z-N) = \frac{1}{2}(C-A) \times (Z-N) = \frac{1}{2}(C-A) \times (Z-A) = -\frac{1}{2}(A-C) \times (Z-A) = -\frac{1}{2}\beta_Z \delta < 0,$$

where $(A - C) \times (Z - A) = (A - C) \times (Z - C) + (A - C) \times (C - A) = \beta_Z \delta$. So $\angle(NC \rightarrow NZ) = -\angle CNZ$. $\square$

Both K and L are on the C -side of AB and on the B -side of AC (interior points; (2.2)). Hence:

$$\angle(BA \rightarrow BK) = -\angle ABK = -\theta, \quad \angle(BA \rightarrow BL) = -\angle ABL, \quad (3.1)$$

$$\angle(CA \rightarrow CL) = +\angle ACL = +\theta, \quad \angle(CA \rightarrow CK) = +\angle ACK, \quad (3.2)$$

$$\angle(MB \rightarrow MK) = +\angle BMK = +\psi, \quad (3.3)$$

$$\angle(NC \rightarrow NL) = -\angle CNL = -\angle LNC = -\varphi. \quad (3.4)$$

Use of “ K inside angle LBA ”. Since L is interior to ABC , L is inside angle ABC (same side of AB as C and of BC as A , by (2.2)); by Lemma 5 applied to angle ABC ($0^\circ < \angle ABC < 180^\circ$ as A, B, C form a triangle), $0^\circ < \angle LBA < \angle ABC < 180^\circ$. Applying Lemma 5 to angle LBA and the point K inside it:

$$\angle ABL = \angle ABK + \angle KBL = \theta + \varphi', \quad \varphi' := \angle LBK.$$

By hypothesis $\varphi' = \angle LNC = \varphi$. Hence, using (0.1) and (3.1),

$$\angle(BK \rightarrow BL) \equiv \angle(BA \rightarrow BL) - \angle(BA \rightarrow BK) = -(\theta + \varphi) + \theta = -\varphi \pmod{360^\circ}. \quad (3.5)$$

Use of “ L inside angle ACK ”. K is interior to ABC , hence inside angle ACB , so $0^\circ < \angle ACK < \angle ACB < 180^\circ$ by Lemma 5. Applying Lemma 5 to angle ACK and the point L inside it: $\angle ACK = \angle ACL + \angle LCK = \theta + \psi'$ with $\psi' := \angle LCK = \angle BMK = \psi$ by hypothesis. Hence by (3.2),

$$\angle(CL \rightarrow CK) \equiv \angle(CA \rightarrow CK) - \angle(CA \rightarrow CL) = (\theta + \psi) - \theta = +\psi \pmod{360^\circ}. \quad (3.6)$$

Two consequences (“nondegeneracy”): since $0 < \varphi < 180^\circ$ and $0 < \psi < 180^\circ$ by (2.3),

$$L \notin \text{line } BK \quad (\text{by (3.5)}), \quad K \notin \text{line } CL \quad (\text{by (3.6)}), \quad (3.7)$$

because if L lay on line BK (note $L \neq B$: L is interior to ABC), the ray BL would coincide with ray BK or its opposite, forcing $\angle(BK \rightarrow BL) \in \{0^\circ, 180^\circ\}$, contradicting (3.5) since $-\varphi \not\equiv 0^\circ, 180^\circ \pmod{360^\circ}$; likewise for K and line CL (note $K \neq C$), using (3.6).

4. The three circles

Recall line $BY = \text{line } BK$ and line $CX = \text{line } CL$.

Claim 4.1 (ω_1). *The points B, C, X, Y lie on one circle ω_1 .*

Proof. The points X, Y, B are pairwise distinct and non-collinear: if B were on line XY then (as $X \neq B$) line $XY = \text{line } XB = AB \ni Y$, contradicting $Y \notin AB$ (indeed Y lies on segment AC with $Y \neq A$, and $AB \cap AC = \{A\}$ because A, B, C are not collinear). Also $C \notin \{X, Y\}$. Now compute directed line angles mod 180° :

$$\angle(BX, BY) \equiv \angle(BA \rightarrow BK) \equiv -\theta \quad (\text{line } BX = AB \text{ since } X \in AB, X \neq B; \text{ line } BY = \text{line } BK),$$

$\angle(CX, CY) \equiv -\angle(CY, CX) \equiv -\angle(CA \rightarrow CL) \equiv -\theta$ (line $CY = AC$ since $Y \in AC$, $Y \neq C$; line $CX =$ line

So $\angle(CX, CY) \equiv \angle(BX, BY) \pmod{180^\circ}$. By Lemma 2 (with $P = X$, $Q = Y$, $R = B$, $S = C$), C lies on the circle ω_1 through X, Y, B . $\square$

Since $X, B \in \omega_1$ are distinct points of line AB , and $Y, C \in \omega_1$ are distinct points of line AC , Lemma 1(i),(iii) give

$$\text{pow}(A, \omega_1) = xc = yb. \quad (4.1)$$

(Signed products: both X and B lie on the ray from A towards B , etc.) Also

$$\text{pow}(B, \omega_1) = 0, \quad \text{pow}(C, \omega_1) = 0, \quad (4.2)$$

and, since $B \neq Y$ are two points of ω_1 on line BK and $C \neq X$ are two points of ω_1 on line CL ,

$$\text{pow}(K, \omega_1) = \langle B - K, Y - K \rangle, \quad \text{pow}(L, \omega_1) = \langle C - L, X - L \rangle. \quad (4.3)$$

Claim 4.2 (ω_K). *Let ω_K be the circumcircle of C, M, K (these are non-collinear: $K \notin$ line CM and $K \notin AB \supseteq \{M\}$ by (P1)). Then line AB meets ω_K exactly in the multiset $\{M, X\}$: if $X \neq M$ then $\omega_K \cap AB = \{M, X\}$; if $X = M$ then AB is tangent to ω_K at M .*

Proof. By (3.3) and (3.6), $\text{mod } 180^\circ$:

$$\angle(CL, CK) \equiv \psi \equiv \angle(MB, MK) = \angle(AB, MK). \quad (4.4)$$

Case $X \neq M$. The points K, X, C are pairwise distinct ($K \notin AB \ni X$; $C \notin AB$; $K \neq C$) and non-collinear: line $XC =$ line CL and $K \notin$ line CL by (3.7). Also $M \notin \{K, X\}$ ($M \in AB$, $K \notin AB$; $X \neq M$ by assumption). Using line $CX =$ line CL and line $MX = AB =$ line MB , (4.4) reads

$$\angle(CK, CX) \equiv -\psi \equiv \angle(MK, MX) \pmod{180^\circ}.$$

By Lemma 2 (with $P = K$, $Q = X$, $R = C$, $S = M$), M lies on the circle through K, X, C ; that circle passes through the non-collinear points C, M, K , hence equals ω_K ; thus $X \in \omega_K$. Since $M \neq X$ are two points of ω_K on AB , Lemma 1(i) gives $\omega_K \cap AB = \{M, X\}$.

Case $X = M$. Then $M \in$ line CL , so (as $C \neq M$) line $CL =$ line CM , and (4.4) becomes $\angle(AB, MK) \equiv \angle(CM, CK) \pmod{180^\circ}$. By Lemma 3(ii) (applied to $\omega_K \ni M, K, C$ and $\ell = AB$), AB is tangent to ω_K at M . $\square$

By Lemma 1(iii) (line AB through A and through B , meeting ω_K at M and X , coincident if $X = M$):

$$\text{pow}(A, \omega_K) = \frac{c}{2}x, \quad \text{pow}(B, \omega_K) = \left(\frac{c}{2} - c\right)(x - c) = \frac{c}{2}(c - x), \quad (4.5)$$

$$\text{pow}(K, \omega_K) = 0, \quad (4.6)$$

and, since $C \neq X$ are two points of ω_K on line CL ,

$$\text{pow}(L, \omega_K) = \langle C - L, X - L \rangle. \quad (4.7)$$

Claim 4.3 (ω_L). *Let ω_L be the circumcircle of B, N, L (non-collinear: $L \notin$ line BN , $L \notin AC \supseteq \{N\}$ by (P2)). Then line AC meets ω_L exactly in the multiset $\{N, Y\}$ (tangency at N if $Y = N$).*

Proof. By (3.4) and (3.5), mod 180° :

$$\angle(BK, BL) \equiv -\varphi \equiv \angle(NC, NL) = \angle(AC, NL). \quad (4.8)$$

Case $Y \neq N$. The points L, Y, B are pairwise distinct and non-collinear (line $BY = \text{line } BK$ and $L \notin \text{line } BK$ by (3.7); $L \notin AC \ni Y$). Also $N \notin \{L, Y, B\}$. Using line $BY = \text{line } BK$ and line $NY = AC = \text{line } NC$, (4.8) reads

$$\angle(BL, BY) \equiv \varphi \equiv \angle(NL, NY) \pmod{180^\circ}.$$

By Lemma 2 (with $P = L$, $Q = Y$, $R = B$, $S = N$), N lies on the circle through L, Y, B , which then equals ω_L ; so $Y \in \omega_L$ and $\omega_L \cap AC = \{N, Y\}$.

Case $Y = N$. Then $N \in \text{line } BK$, so line $BK = \text{line } BN$, and (4.8) becomes $\angle(AC, NL) \equiv \angle(BN, BL) \pmod{180^\circ}$. By Lemma 3(ii) (with the circle $\omega_L \ni N, L, B$ and $\ell = AC$), AC is tangent to ω_L at N . $\square$

Consequently

$$\text{pow}(A, \omega_L) = \frac{b}{2}y, \quad \text{pow}(C, \omega_L) = \frac{b}{2}(b-y), \quad \text{pow}(L, \omega_L) = 0, \quad \text{pow}(B, \omega_L) = 0, \quad (4.9)$$

and, since $B \neq Y$ are two points of ω_L on line BK ,

$$\text{pow}(K, \omega_L) = \langle B - K, Y - K \rangle. \quad (4.10)$$

5. The affine power identity

Define $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(Z) := \text{pow}(Z, \omega) - \text{pow}(Z, \omega_K) - \text{pow}(Z, \omega_L) + \text{pow}(Z, \omega_1).$$

F is affine. For any circle Γ with centre Z_Γ and radius ρ_Γ , $\text{pow}(Z, \Gamma) = |Z|^2 - 2\langle Z_\Gamma, Z \rangle + |Z_\Gamma|^2 - \rho_\Gamma^2$. In F the $|Z|^2$ -terms cancel ($+1 - 1 - 1 + 1 = 0$), so $F(Z) = \langle u, Z \rangle + v$ for a fixed vector u and scalar v .

$F(A) = 0$: by $\text{pow}(A, \omega) = 0$, (4.5), (4.9), (4.1):

$$F(A) = 0 - \frac{c}{2}x - \frac{b}{2}y + cx = \frac{cx}{2} - \frac{by}{2} = 0,$$

using $cx = by$ from (4.1).

$F(K) = 0$: $\text{pow}(K, \omega) = 0$ ($K \in \omega$), $\text{pow}(K, \omega_K) = 0$, and by (4.3), (4.10) $\text{pow}(K, \omega_1) = \text{pow}(K, \omega_L) = \langle B - K, Y - K \rangle$; the last two cancel.

$F(L) = 0$: $\text{pow}(L, \omega) = 0$, $\text{pow}(L, \omega_L) = 0$, and by (4.3), (4.7) $\text{pow}(L, \omega_1) = \text{pow}(L, \omega_K) = \langle C - L, X - L \rangle$.

Hence $F \equiv 0$: A, K, L are not collinear, so $K - A$ and $L - A$ are linearly independent; from $F(A) = F(K) = F(L) = 0$ we get $\langle u, K - A \rangle = \langle u, L - A \rangle = 0$, so $u = 0$, and then $v = F(A) = 0$.

Evaluating F at B and C and using (4.2), (4.5), (4.9):

$$\text{pow}(B, \omega) = \text{pow}(B, \omega_K) + \text{pow}(B, \omega_L) - \text{pow}(B, \omega_1) = \frac{c}{2}(c - x), \quad (5.1)$$

$$\text{pow}(C, \omega) = \text{pow}(C, \omega_K) + \text{pow}(C, \omega_L) - \text{pow}(C, \omega_1) = \frac{b}{2}(b - y). \quad (5.2)$$

6. Conclusion

O is the centre of ω and $R_\omega = |OA|$ its radius, so $|OB|^2 = \text{pow}(B, \omega) + R_\omega^2$ and $|OC|^2 = \text{pow}(C, \omega) + R_\omega^2$. By Lemma 4 applied to the midpoints M of AB and N of AC :

$$4OM^2 = 2OA^2 + 2OB^2 - c^2, \quad 4ON^2 = 2OA^2 + 2OC^2 - b^2.$$

Subtracting and using (5.1), (5.2) and $cx = by$ from (4.1):

$$\begin{aligned} 4(OM^2 - ON^2) &= 2(OB^2 - OC^2) - c^2 + b^2 = 2[\text{pow}(B, \omega) - \text{pow}(C, \omega)] - c^2 + b^2 \\ &= c(c - x) - b(b - y) - c^2 + b^2 = -cx + by = 0. \end{aligned}$$

Therefore $OM = ON$. ■

Remarks on hypothesis usage

- $\angle KBA = \angle ACL \rightarrow$ Claim 4.1; $\angle LCK = \angle BMK$ together with “ L inside angle ACK ” $\rightarrow$ Claim 4.2; $\angle LBK = \angle LNC$ together with “ K inside angle LBA ” $\rightarrow$ Claim 4.3.
- “ K inside triangle BMC ” is used exactly for: K interior to ABC (sign facts (S1)–(S2), crossbar (P3)) and $K \notin$ line CM (ω_K nondegenerate). Similarly for L and line BN .
- That M, N are midpoints enters only through $AM = MB = \frac{c}{2}$, $AN = NC = \frac{b}{2}$ in (4.5), (4.9) and through Lemma 4.
- Numerical verification of every claim (generic and both tangency cases): `code/verify.py`, `code/check_signs.py`, `code/verify2.py`.